9/1/26
PHYS 2500

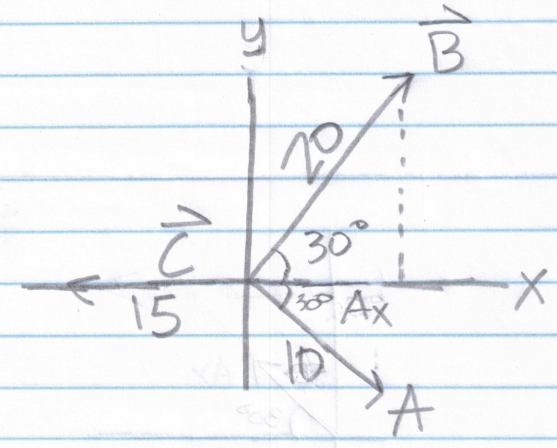
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Worksheet 1

1. Three Vectors are oriented where

$A=10$, $B=20$, $C=15$ find:

a. the X and y Components of the resultant vector $\vec{D} = \vec{A}+\vec{B}+\vec{C}$



$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$

$\text{adjacent} = \frac{\text{opposite}}{\tan(\theta)}$

$= \frac{20}{\tan(30^\circ)}$

$= 34.64$

$\sin\theta = \frac{\text{opp}}{\text{hyp}}$

$= \frac{By}{B}$

$B\sin\theta = By$

$20\sin(30^\circ) = 10$

$20\cos(30^\circ) = 17.3$

$\sin\theta = \frac{\text{opp}}{\text{hyp}}$

$\cos\theta = \frac{\text{adj}}{\text{hyp}}$

$\vec{B} = 17.3\hat{x} + 10\hat{y}$

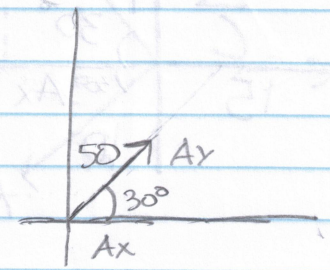
$\vec{C} = -15\hat{x} + 0\hat{y}$

$\vec{A} =$

$\vec{A} = 50N$

$\vec{B} = 100N$

$\vec{C} = 50N$



$A_x = ?$ $\quad 50 \sin(30°) = 25°$

$A_y = ?$ $\quad 50 \cos(30°) = 43.3°$

$B_x = ?$ $\quad 100 \sin(30°) = -50°$

$B_y = ?$ $\quad 100 \cos(30°) = 86.6°$

$C_x = ?$

$C_y = ?$